

Automated Network Request Management

Built on the ServiceNow Service Catalog

1. Introduction

Item	Details
Project Title	Automated Network Request Management
Platform	ServiceNow (developer instance), Service Catalog module
Catalog Item	Network Request (Category: Network Standard Changes)
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2. Project Overview

Purpose

Network changes are often requested through emails and tickets, which is slow and easy to get wrong. This project replaces that with a single Service Catalog item, **Network Request**, that collects the right information from the requester, validates it on the form, and routes it through the ServiceNow process engine so that network standard changes are handled in a consistent, trackable way.

Key Features

- **Catalog item:** one 'Network Request' item (short description: Network Services Request) published in the Service Catalog under 'Network Standard Changes'.
- **Variables:** seven catalog variables capture the request details.
- **Reusable variable set:** the 'Requester Information' set (2-column layout, 4 variables, including 'opened on behalf of') captures who the request is for.
- **Catalog UI Policy:** 'visible/hide of ID field' shows or hides the ID field depending on the form state (Reverse if false: true, On load: true).
- **Catalog Client Scripts:** onLoad / onChange scripts populate and control fields, for example 'Populate Name with field' and 'onChange of Mandatory Attachment'.
- **Process engine:** request fulfilment is handled by the platform process engine and the item's Process Engine tab.
- **Try It:** the item can be tested straight from the catalog item form.

3. Architecture

ServiceNow is a low-code platform, so the MERN layers in the standard template map onto ServiceNow components as shown below.

Layer	MERN equivalent	Implementation in this project
Frontend	React UI	Service Catalog item form (Service Portal / Platform UI) with variables, a variable set, Catalog UI Policy and Catalog Client Scripts running in the browser.
Backend	Node.js + Express	ServiceNow server: catalog item process engine, request and requested-item records, business logic and flows.
Database	MongoDB	ServiceNow relational tables such as <code>sc_cat_item</code> , <code>catalog_script_client</code> , the custom table <code>u_database_tables</code> , plus the request tables created on submission.

Frontend

- The Network Request form is made up of the Requester Information variable set (layout: 2 columns wide, one side, then the other) and the item's own variables.
- Catalog UI Policy** 'visible/hide of ID field' controls field visibility without code.
- Catalog Client Scripts** (type onLoad, onChange, onSubmit) run in the browser to fill fields, enforce mandatory attachments and hide variables.

Backend

- The **Process Engine** tab on the catalog item defines how the request is processed after submission.
- Fulfilment automation level is currently *Unspecified*; item owner is System Administrator.

Database

- Catalog items live in `sc_cat_item`; client scripts in `catalog_script_client`; variable sets and variables in the platform's variable tables.
- The custom table **Database Tables** (`u_database_tables`, application: Global, 16 columns) was created for the project. Columns seen in the video include Mobile Number (String, max length 40), Updated (Date/Time) and Mode of Payment (String, max length 40).
- Table creation options enabled: *Create module* and *Create mobile module*.

4. Setup Instructions

Prerequisites

- A ServiceNow instance (a free Personal Developer Instance from developer.servicenow.com is enough).
- An account with the **admin** role (needed to create tables, catalog items and scripts).
- A modern web browser (Chrome, Edge or Firefox).

Installation / Configuration Steps

- Step 1 - Create the table:** go to *System Definition > Tables > New*. Set Label = Database Tables, Name = `u_database_tables`, Application = Global, and add the required columns.
- Step 2 - Create the variable set:** go to *Service Catalog > Catalog Variables > Variable Sets*. Create 'Requester Information' (internal name `requester_information`, Order 100, Type: Single Row, Layout: 2 Columns Wide) and add its variables, including 'opened on behalf of' (Reference).
- Step 3 - Create the catalog item:** go to *Service Catalog > Catalog Definitions > Maintain Items > New*. Name = Network Request, Catalogs = Service Catalog, Category = Network Standard Changes, Short description = Network Services Request, Active = true.

- **Step 4 - Add variables:** on the item, open the Variables related list and add the seven variables; attach the Requester Information variable set.
- **Step 5 - Add the UI Policy:** in the Catalog UI Policies related list, create 'visible/hide of ID field' (Reverse if false, On load, Order 100).
- **Step 6 - Add Client Scripts:** in the Catalog Client Scripts related list, add the onLoad and onChange scripts required by the form.
- **Step 7 - Test:** click **Try It** on the catalog item and submit a sample request.

Environment settings (equivalent of environment variables): instance URL, admin user, and the update set / application scope (Global in this project).

5. Folder Structure

ServiceNow has no folder tree on disk. The equivalent structure is the way the artefacts are organised inside the platform:

```
Service Catalog (Catalog)
+-- Network Standard Changes (Category)
    +-- Network Request (Catalog Item)
        +-- Item Details / Process Engine / Picture / Pricing / Portal Settings
        +-- Variables (7)
        +-- Variable Sets (1) ... Requester Information (4 variables)
        +-- Catalog UI Policies (1) ... visible/hide of ID field
        +-- Catalog Client Scripts ... onLoad / onChange / onSubmit
        +-- Available For / Not Available For (access control)
        +-- Categories (1), Catalogs (1)
```

```
Tables
+-- u_database_tables (Database Tables, 16 columns)
```

Client (browser side): catalog variables, variable sets, Catalog UI Policies and Catalog Client Scripts.
Server side: process engine, request and requested-item records, and server business logic.

6. Running the Application

There are no npm start commands. The application runs inside the ServiceNow instance:

- Log in to the instance and open **All > Service Catalog > Catalog Definitions > Maintain Items**.
- Search for **Network Request** and open the record.
- Click **Try It** to open the form as an end user, fill in the fields and submit.
- End users reach the same item through the Service Catalog / Service Portal under Network Standard Changes.

7. API Documentation

The project itself is configured in the UI. The standard ServiceNow REST APIs below can be used to read or create the same data (these are platform endpoints; they were not shown in the video). Replace {instance} with your instance name.

Method	Endpoint	Purpose
GET	/api/now/table/sc_cat_item?sysparm_query=name=Network Request	Fetch the Network Request catalog item
GET	/api/now/table/catalog_script_client?sysparm_query=cat_item={sys_id}	List client scripts of the item

Method	Endpoint	Purpose
GET	/api/now/table/u_database_tables	List records of the custom table
POST	/api/now/table/u_database_tables	Create a record in the custom table
POST	/api/sn_sc/servicecatalog/items/{sys_id}/order_now	Order the catalog item with variable values

Example request and response

POST https://{instance}.service-now.com/api/sn_sc/servicecatalog/items/{sys_id}/order_now
Content-Type: application/json

```
{ "sysparm_quantity": "1",
  "variables": { "opened_on_behalf_of": "<user sys_id>" } }
```

200 OK

```
{ "result": { "sys_id": "...", "number": "REQ0010001",
              "request_number": "REQ0010001", "table": "sc_request" } }
```

Variable names and the sample response are illustrative; confirm the real variable names on the item's Variables tab.

8. Authentication

- **User login:** users sign in to the instance with a ServiceNow username and password (or SSO if configured). A session cookie is used for the browser UI.
- **Authorisation:** access is controlled by **roles** and **ACLs**. The catalog item has a Roles field and 'Available For / Not Available For' lists to restrict who can see and order it. Development tasks need the admin role.
- **REST API access:** Basic Authentication or OAuth 2.0 can be used with the endpoints in section 7, with the same role checks applied.

9. User Interface

Key screens of the solution, taken from the project walkthrough video.

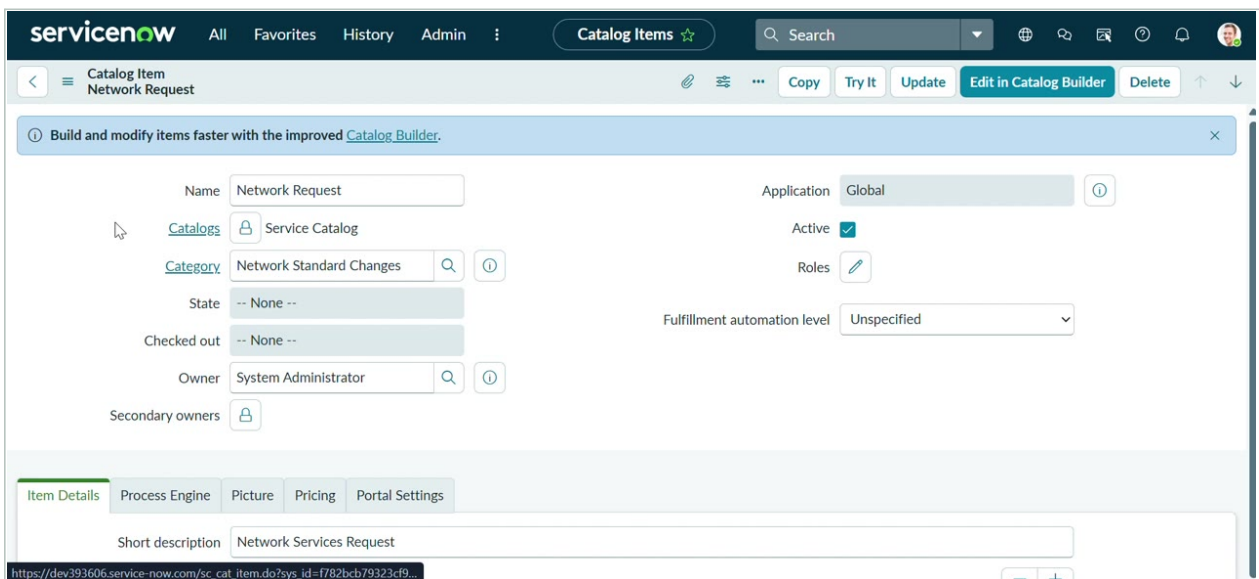


Figure 1 - Network Request catalog item (Maintain Items): Service Catalog, category Network Standard Changes.

servicenow All Favorites History : Catalog Item - Network Request Search

< Catalog Item Network Request Copy Try It Update Edit in Catalog Builder Delete

Copy Try It Update Edit in Catalog Builder Delete

Related Links
[Item Diagnostic](#)
[Show VA render type](#)
[Run Point Scan](#)

Variables (7) Variable Sets (1) **Catalog UI Policies (1)** Catalog Client Scripts Available For Not Available For Categories (1) Catalogs (1) Catalog Data Lookup Definitions

Related Articles Related Catalog Items **Assigned Topics**

Order Search Actions on selected rows... New

Catalog Item = Network Request

	Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
☐	visible/hide of ID field	(empty)		true	true	false	2026-10-02 02:37:44	100

1 to 1 of 1

Figure 2 - Catalog UI Policy 'visible/hide of ID field' in the item's related lists.

servicenow All Favorites History : Variable Set - Requester Information Search

< Variable Set Requester Information Update Delete

* Title **Requester Information** Application Global

* Internal name requester_information Display title ☒

Order 100 Layout 2 Columns Wide, one side, then tr

Type Single Row

Description

Update Delete

Variables (4) **Catalog UI Policies** Catalog Client Scripts Included In (1) Catalog Data Lookup Definitions

Order Search Actions on selected rows... New

Variable set = Requester Information

	Name	Type	Question	Order
☐	opened on behalf of	Reference	Opened on behalf of	

Figure 3 - Variable Set 'Requester Information' (2 columns wide, 4 variables).

servicenow All Favorites History Admin : Catalog Client Scripts Search

< Catalog Client Scripts Name Search Actions on selected rows... New

All > Name >= network request

	Name	Active	Table	View	Type	Updated
☐	Search	Search	Search	Search	Search	Search
	onChange of Hide Mandatory	true			onChange	2022-11-20 06:07:10
	onChange of Mandatory Attachment	true			onChange	2022-11-20 06:07:21
	onChange Process engine	true			onChange	2022-11-20 06:04:05
	onLoad Process engine	true			onLoad	2022-11-28 01:09:52
	Populate applies to Item field	true			onLoad	2025-08-10 22:30:55
	Populate condition variable	true			onLoad	2021-08-03 01:20:41
	Populate Is Email & SMS active on load	true			onLoad	2023-03-22 12:24:15
	Populate Name with field	true			onChange	2025-04-13 03:18:08
	Populate Name with question text	true			onChange	2022-11-01 11:21:33
	Populate Prediction fields on load	true			onLoad	2022-06-03 08:08:09
	Populate Roles and Hide Variables	true			onLoad	2010-03-16 08:05:33

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Figure 4 - Catalog Client Scripts filtered for the Network Request item.

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label: Database Tables
 * Name: u_database_tables
 Application: Global
 Remote Table: ☐

Columns | Controls | Application Access

Table Columns: for text Search

Dictionary Entries

	Column label	Type	Reference	Max length	Default value	Display
×	Mobile Number	String	(empty)	40		false
	Updated	Date/Time	(empty)	40		false
×	Mode of Payment	String	(empty)	40		false

Figure 5 - Custom table 'Database Tables' (u_database_tables) with its columns.

10. Testing

- **Try It:** run the catalog item from the form to test variables, UI policy and client scripts as an end user.
- **Functional checks:** verify mandatory fields, the show/hide behaviour of the ID field, auto-populated fields, and that attachment rules fire on submit.
- **Process checks:** confirm a request and requested item are created and move through the process engine.
- **Role testing:** impersonate a normal user to confirm access limits.
- **Optional automation:** ServiceNow Automated Test Framework (ATF) can be used for regression tests.

11. Screenshots or Demo

Walkthrough video: *Automated Network Request Management* (about 3 min 27 s). Additional screens:

	Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
	3M Privacy Filter - Lenovo X1 Carbon	Privacy Filter - X1 Carbon	true		Service Catalog	Peripherals	\$43.19	Item	2022-11-20 20:46:33
	3M Privacy Filter - MacBook Pro	Privacy Filter	true		Service Catalog	Peripherals	\$42.23	Item	2022-11-20 20:46:33
	3M Privacy Filter - Macbook Pro Retina	Privacy Filter	true		Service Catalog	Peripherals	\$40.31	Item	2022-11-20 16:00:00
	Access	Microsoft Access	true		Service Catalog	Software	\$139.99	Item	2022-11-20 20:46:33
	Acrobat	Adobe Acrobat	true		Service Catalog	Software	\$139.99	Item	2022-12-05 20:46:33
	Add network switch to datacenter cabinet	This standard change template describes ...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-09-13 17:24:23
	Add/Remove users from group	Add/Remove users from group	true		Service Catalog	Services	\$0.00	Item	2022-12-05 10:17:33
	Adobe Acrobat Pro	Create, edit or convert PDF files	true		Service Catalog	Software	\$0.00	Item	2022-12-05 20:46:33

Figure 6 - Catalog Items list filtered by 'network'.

Name	Active	Table	View	Type	Updated
onChange of Hide Mandatory	true			onChange	2022-11-20 06:07:10
onChange of Mandatory Attachment	true			onChange	2022-11-20 06:07:21
onChange Process engine	true			onChange	2022-11-20 06:04:05
onLoad Process engine	true			onLoad	2022-11-28 01:09:52
Populate applies to Item field	true			onLoad	2025-08-10 22:30:55
Populate condition variable	true			onLoad	2021-08-03 01:20:41
Populate Is Email & SMS active on load	true			onLoad	2023-03-22 12:24:15
Populate Name with field	true			onChange	2025-04-13 03:18:08
Populate Name with question text	true			onChange	2022-11-01 11:21:33
Populate Prediction fields on load	true			onLoad	2022-06-03 08:08:09
Populate Roles and Hide Variables	true			onLoad	2010-03-16 08:05:33

Figure 7 - 81 Catalog Client Scripts matching the Network Request item.

12. Known Issues

- Fulfilment automation level is still 'Unspecified', so no automated fulfilment is configured yet.
- The item's description field is empty; users get little guidance on the form.
- Several catalog client scripts shown are older platform scripts (2010-2023); they should be reviewed for conflicts.
- The video shows the Catalog Client Scripts list for 'Catalog item = Network Request' empty at first, so item-specific scripts need to be confirmed.
- [Add any bugs you have found during testing.]

13. Future Enhancements

- Add approval flows (e.g. network manager approval) and automated fulfilment with Flow Designer.
- Integrate with network or CMDB tools to apply or validate changes automatically.
- Add notifications (email / SMS) for status updates.
- Build dashboards and reports for request volumes and turnaround time.
- Provide a mobile-friendly Service Portal widget and an AI-search friendly item description and meta tags.
- Write ATF tests so changes can be regression-tested.